PRUDHVI REDDY

Senior AI Engineer

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SUMMARY

AI engineer with 6 years of experience in machine learning and building LLM applications that do real work, agents that plan multistep tasks, call the right tools, and hand decisions back to people when the stakes are high. At Sam's Club, leads the team behind an operations agent platform where multiple agents share context through MCP, ground their answers in company knowledge, and now resolve routine supplier and inventory exceptions without manual effort. Earlier work on a medical device knowledge assistant for Los Angeles County hospitals and clinical AI at Mayo Clinic built a careful habit of evaluating models, protecting patient data, and earning the trust of clinical users. Keeps close watch on new models and agent frameworks, and adopts them only after testing them against real use cases.

CORE SKILLS

Languages:	Python, SQL, TypeScript, C#.Net, Java
LLMs:	OpenAI GPT-4o, Anthropic Claude, Azure OpenAI, Llama 3, Hugging Face Transformers, vLLM
Agent Frameworks:	LangGraph, LangChain, AutoGen, CrewAI, Multi agent orchestration, Planning and tool calling, Human in the loop
Model Context Protocol:	MCP servers and clients, Tools and resources, Persistent session context, Agent interoperability
RAG and Retrieval:	Pinecone, Milvus, ChromaDB, FAISS, Embeddings, Hybrid search, Reranking, Chunking strategies
ML and Fine Tuning:	PyTorch, TensorFlow, LoRA, DPO, Prompt engineering, Structured outputs, MLflow
Backend and Cloud:	FastAPI, Flask, Docker, Kubernetes (AKS, EKS), Azure, AWS, Git, GitHub Actions
Observability and Safety:	LangSmith, OpenTelemetry, Grafana, LLM evaluation, Guardrails, PII redaction, Prompt injection defense, Responsible AI

Tools | GitHub Actions, Terraform, Jupyter, VS Code, JIRA, Confluence

PROFESSIONAL EXPERIENCE

Senior AI Engineer| Walmart (Client) | Los Angeles, CA **Sep 2025 – Present**

- Led a team of four building the Sam's Club operations agent platform, where LLM agents resolve supplier delivery exceptions and club inventory issues across about 600 clubs by reasoning over orders, stock, and supplier data and acting through internal APIs.
- Architected multi agent workflows in LangGraph, with a planner agent that breaks each request into steps and specialist agents for inventory, orders, and supplier messages, chosen over AutoGen and CrewAI after a prototype comparison.
- Built MCP servers in Python that expose inventory, purchase order, and ticketing systems as typed tools and resources, giving every agent one governed interface with persistent session context and full audit logs.
- Developed the RAG layer over supplier contracts, procedures, and past exception cases using hybrid search in Pinecone, Azure OpenAI embeddings, and a reranker, raising answer groundedness from 71% to 92% on the evaluation set.
- Adapted Llama 3 8B through LoRA fine tuning in PyTorch and Hugging Face for exception routing, then aligned it with DPO on about 8,000 buyer approvals and rejections, matching GPT-4o accuracy at roughly a fifth of the cost on vLLM.
- Deployed agent services behind FastAPI with streaming responses, in Docker on AKS with autoscaling, released through GitHub Actions with prompts and tool schemas versioned in Git.
- Instrumented every agent run with LangSmith and OpenTelemetry tracing, Grafana dashboards for latency, token use, and cost, and offline evaluation suites that gate each prompt or model change.
- Enforced responsible AI controls such as PII redaction, prompt injection filters, scoped tool permissions, and human approval before any order change, reviewed with security and privacy teams.

AI Engineer| Los Angeles County (Client) | Los Angeles, CA **Mar 2024 – Sep 2025**

- Delivered an LLM assistant for Department of Health Services biomedical engineering teams that answers questions on medical device manuals, service procedures, and FDA recall notices for equipment across county hospitals.
- Engineered the RAG pipeline in LangChain with layout aware PDF parsing, section based chunking, and embeddings stored in Milvus, returning cited answers linked to the source page.
- Created an AutoGen agent pair that matches new FDA recall notices against the county device inventory through an MCP server and drafts work orders for affected equipment, cutting recall review from days to hours.
- Designed prompts with structured JSON outputs and few shot examples, testing each variant against a 300 question evaluation set written with clinical engineers.

- Benchmarked GPT-4o, Claude, and open Hugging Face models on accuracy, latency, and cost, then routed simple lookups to a smaller model and cut inference spend by about 45%.
- Shipped the services with FastAPI and Docker on Amazon EKS, keeping all device and patient related data inside the county's AWS environment under HIPAA controls.
- Monitored production use through logged prompts, responses, user feedback, and hallucination flags reviewed weekly with the clinical engineering lead.

Software Engineer | Mayo Clinic (Client) | [Remote **May 2023 – Dec 2023**

- Prototyped LLM clinical note summarization with GPT-4 through Azure OpenAI and LangChain, cutting chart review from about 25 minutes to 8 per patient in a pilot with 40 care managers.
- Grounded summaries in care guidelines using FAISS and sentence transformer embeddings, so each summary cited the protocol it relied on.
- Fine tuned a clinical BERT model in PyTorch to flag patients who needed follow up from free text notes.
- Served the models through Flask APIs in Docker containers integrated with care management applications.
- Validated outputs with nurse reviewers on 200 charts, tracking hallucination and omission rates before release.
- Protected patient data with de identification, least privilege access, and audit logging, passing an internal HIPAA security review.

Machine Learning Engineer| Dell Technologies (Client) | India **Jan 2019 to Dec 2021**

- Forecast hardware component failures with TensorFlow models trained on device telemetry, used to open proactive support cases.
- Constructed feature pipelines in Python and Spark over telemetry from customer laptops and servers.
- Classified support case notes with an NLP model that routes cases to the right product team.
- Packaged models as Flask services in Docker with alerts for prediction drift.
- Tracked experiments and model versions in MLflow and Git.

PROJECTS

MCP Agent Toolkit (open source) | Python, MCP, LangGraph, ChromaDB

- Built a reference MCP server that exposes a ChromaDB knowledge store and a ticketing API as tools, with a LangGraph client showing multi agent planning and shared context.

Multi-Agent Support Triage | LangGraph, LangChain, RAG, FastAPI, React, Redis, Kubernetes

- Built a full-stack support triage application with a React front end and FastAPI back end, deployed with Docker and Kubernetes through GitHub Actions CI/CD, with Redis caching and LangSmith tracing.

EDUCATION

Master of Science (M.S.), Computer Science | University of Central Missouri | Jan 2022 – Dec 2023

Bachelor of Science (B.S.), Computer Science | Jawaharlal Nehru Technological University | Jun 2017 – Aug 2021

CERTIFICATIONS

Microsoft Certified Azure AI Engineer Associate (AI 102) | AWS Certified Machine Learning Engineer Associate